Five-Model Comparison

Two baselines, the proposed model, the caution-aware model and an image-only model: how each was built, trained, tested and what it achieved

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1 What this document is

This document is for the people working on the project. It compares **five driving models**, one section each, with the same headings every time: what it sees, the network, how it was trained, how it was tested, how its energy is computed, what it achieved, which graph shows it, and *what we are allowed to say about it.* A shorter story of the project is in the *Director's Research Guide*; every detail, every experiment and every file is in the *Complete Technical Research Report.*

The five models in one table

All five models take the same job: drive a car from a start to a goal in the MetaDrive simulator, steering and using throttle or brake. They differ in *what they look at* and *how they were taught.* Four of them look at a small top-down picture (a "bird's-eye-view grid"); one of them looks at the raw front-camera picture.

Table 1: The five models. "Teacher" is the rule-based driver whose actions the model copies. Models 1 and 2 were built by the teammates and are on GitHub; models 3 to 5 exist only on the local machine.

	Model	What it looks at	How it was taught	Role in the comparison
1	Baseline Driving Model Trained Without Traffic	Bird's-eye-view grid, six vehicle-state numbers	Copy of 5 recorded drives, no other vehicles	Teammates' baseline (weaker)
2	Baseline Driving Model Trained With Traffic	Same	Copy of 5 recorded drives, sparse traffic	Teammates' baseline, supposedly stronger
3	Proposed Bird's-Eye-View Driving Model	Grid with the road drawn in, three vehicle-state numbers	Repeated drive-and-correct (dataset aggregation)	The strong method
4	Caution-Study Caution-Aware Driving Model	As model 3 plus one "caution" number	As model 3 but the teacher drives slower and keeps more distance when caution is high	Adaptive-behaviour candidate
5	Image-Only Driving Model (front camera)	Only the front-camera picture	Repeated drive-and-correct on camera images	The weak control: what you get without the grid

Strong against weak. Model 5 is deliberately the weakest design: no grid, no road drawing, no speed input. It shows what the other four gain from the bird's-eye-view grid. Models 1 and 2 are the teammates' starting point. Model 3 is the strong method. Model 4 asks whether the strong method can also adapt to weather.

Plain-language rule used throughout. Every model is called by its descriptive name. The code nicknames (for example `ov_split`) appear only in the file list at the end.

2 How every model is tested (the same way for all five)

2.1 The test

- **Simulator.** MetaDrive 0.4.3. A *scenario seed* (a whole number) fixes the map, the route and the starting traffic, so every model drives exactly the same 200 scenarios (seeds 5000 to 5199). These seeds were never used for training or for choosing a model.

- **Traffic.** Sparse: traffic density 0.05. **Time limit:** 800 steps of 0.1 s = 80 seconds (a 160-second check is in Section 7.7).
- **An episode ends** with exactly one of four outcomes: *success* (destination reached), *collision*, *left the road*, or *timeout*.
- **Weather conditions:** clear, fog 1.0, rain 0.6, rain 1.0 (the number is the intensity of the teammates' weather filter; 1.0 is the strongest).

2.2 How the weather reaches each model

Weather never changes the road grip or the car's physics. It only damages perception.

Models 1 to 4 (grid models)
They do not look at camera pictures in these tests. The grid is built from the simulator's list of vehicles, and *each vehicle is removed from the grid with the probability that the detector was measured to miss it* at that weather intensity. That measurement is the detector retention in Figure 1. So the weather reaches these models *indirectly*, through a measured number, not through pictures.

Model 5 (image-only)
It looks at the real camera picture, and the teammates' weather filter (fog haze, rain streaks) is applied to that picture directly. This is the strongest form of weather test, but it is applied to a different kind of input.

Honest limit

Models 1 to 4 are tested with *measured* detector retention but with an ideal detector otherwise (the grid is built from the simulator's vehicle list). The GitHub README asks that the policy never receives exact simulator positions; these tests depart from that rule, and the result says how the models behave under a measured loss of vehicles, not how a full camera system would behave.

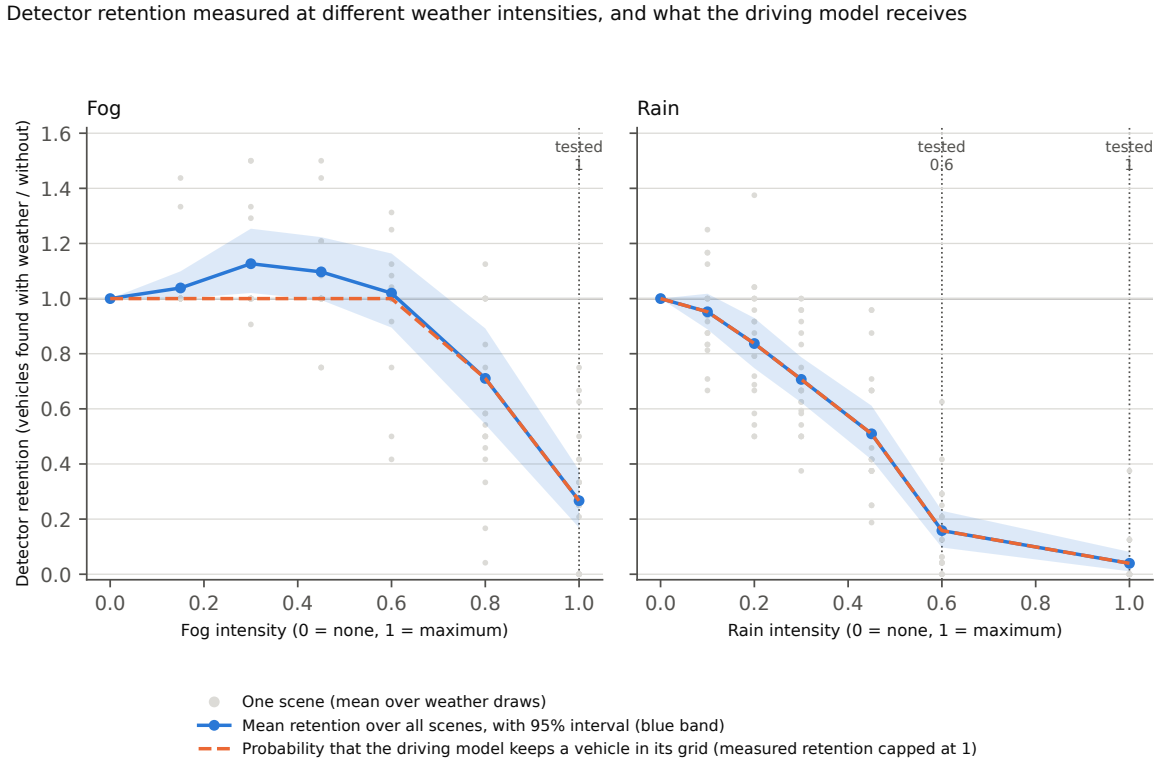


Figure 1: Detector retention measured at different fog and rain intensities, and the probability that the driving model keeps a vehicle. In 20 scenes (seeds 1000 to 1019, traffic density 0.3, stationary car, up to three vehicles placed around it) the same camera frames were passed through the teammates' vehicle detector twice: once clean and once with the weather filter at each intensity (two random weather draws per scene). Retention is the number of vehicles found with weather divided by the number found without weather. Grey dots are single scenes, the blue line is the mean over scenes with its 95% interval, the orange dashed line is the measured retention capped at 1, which is the probability that the weather proxy keeps a vehicle in the driving model's grid. Dotted vertical lines mark the intensities used in the 200-episode weather tests.

Plain-English reading. Fog does almost nothing up to intensity 0.6, then removes vehicles quickly: about 71% are kept at 0.8 and 27% at 1.0. Rain acts earlier and harder: about half of the vehicles are kept at 0.45, 16% at 0.6 and 4% at 1.0. Values slightly above 1 in light fog mean that the detector found a few more boxes than in the clean frame; they are treated as no loss.

Research implication. The driving model never sees the weather and never sees camera images in these tests. It receives the weather only indirectly: each vehicle is removed from its bird's-eye-view grid with the probability read off this curve. The conclusions about weather therefore hold for *loss of detected vehicles at these measured rates*, from 20 scenes per point; they do not cover other perception failures.

2.3 The numbers reported, and their formulas

Every symbol, the reason for each formula, its derivation and its source are explained in Section ef-sec:formulas; this subsection gives the short version. Let $n = 200$ be the number of episodes and k the number that ended in success.

Success rate $\hat{p} = k/n$. Collision, left-road and timeout rates are computed the same way; the four rates add up to 1.

95% interval (Wilson)
$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$, $z = 1.96$. It is the range of true success rates that is compatible with the 200 episodes we saw.

Paired comparison (exact McNemar test)
Two models are compared on the same seeds. Let b be the number of episodes only model A solved and c the number only model B solved. Under "no difference" each of these $b + c$ episodes is equally likely to go either way, so $p = \min\Big(1,\ 2\sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)}\Big)$. A small p means the difference is unlikely to be chance.

Mean speed Average of the car's speed over all steps of an episode (m/s), then averaged over the 200 episodes.

Route completion
Fraction of the planned route driven when the episode ended (1.0 = the destination).

Estimated energy
Same formula for every model, applied to the speed the model produced (below).

2.4 How energy is computed (identical for all five models)

MetaDrive has no electric-energy model. We therefore estimate the *mechanical work at the wheels* of a declared virtual car, using only the speed history of each episode. The model does not influence the formula; a model uses less energy only because it drives differently (mostly slower or smoother). For one step of $\Delta t = 0.1\,\text{s}$ with start speed v_0 and end speed v_1 (speed assumed to change in a straight line inside the step):

$$
\begin{aligned}
W_{\text{inertia}} &= \tfrac{1}{2}\, m\,(v_1^2 - v_0^2) && \text{(change of kinetic energy)}\\
W_{\text{air}} &= k\,\Delta t\,\tfrac{1}{4}\,(v_0^3 + v_0^2 v_1 + v_0 v_1^2 + v_1^3), \quad k = \tfrac{1}{2}\rho C_d A && \text{(air drag)}\\
W_{\text{roll}} &= C_{rr}\, m\, g\, \tfrac{v_0+v_1}{2}\,\Delta t && \text{(tyre rolling resistance)}\\
W_{\text{step}} &= \max\left(0,\ W_{\text{inertia}} + W_{\text{air}} + W_{\text{roll}}\right)
\end{aligned}
$$

with air density $\rho = 1.225\,\text{kg/m}^3$, drag coefficient $C_d = 0.23$, frontal area $A = 2.22\,\text{m}^2$, rolling resistance $C_{rr} = 0.01$, gravity $g = 9.81\,\text{m/s}^2$ and the car mass m taken from the simulator's vehicle. Then

$$
E_{\text{Wh}} = \frac{\sum_{\text{steps}} W_{\text{step}}}{3600}, \qquad D_{\text{km}} = \frac{1}{1000}\sum_{\text{steps}} \tfrac{v_0+v_1}{2}\Delta t, \qquad \text{energy per kilometre} = \frac{E_{\text{Wh}}}{D_{\text{km}}}.
$$

Braking steps count as zero (the clip at 0): there is *no regenerative braking*, so braking energy is lost, not recovered. Road slope is zero (the simulator's roads are flat). **Only differences between models are meaningful; this is not a battery measurement.** The reported value of a model is the mean over its 200 episodes of the per-episode energy per kilometre.

Table 2: Energy computation per model. The formula is the same; only the speed history differs.

Model	What determines its energy
1, 2 Baselines	Their own speed history: they hold about 8.1 m/s and brake little.
3 Proposed	Its own speed history: about 7.9 m/s; slightly more braking and accelerating than the baselines, hence slightly more energy.
4 Caution-aware	Slower in fog and rain (5.6 to 6.2 m/s), which lowers drag work strongly (drag grows with the cube of speed).
5 Image-only	Fast and erratic (about 9 m/s), with many accelerations; it uses much more energy per kilometre, but most episodes end early by leaving the road.

3 Formula guide: what every symbol means, why we use it, where it comes from

In plain words

This section explains every formula of the document in the same order: **(1) what it computes, (2) the meaning of each symbol, (3) why this formula and not another, (4) how it is derived, (5) where it comes from.** Links point to the arXiv page of the paper. Some classical sources (1927 to 1947) have no arXiv version; for those the journal reference is given. Anything that is our own choice, not taken from the literature, is marked **our choice**.

3.1 Success rate and the other outcome rates

$$\hat{p} = \frac{k}{n}$$

Symbols n: number of episodes (200). k: how many of them ended in the outcome being counted (success, collision, left the road or timeout). $\hat{p}$: the observed share (the "hat" means "estimated from data"). Every episode ends in exactly one of the four outcomes, so the four shares add up to 1.

Why It is the simplest honest summary of closed-loop driving: "out of the scenarios we gave it, how many did it finish". We do not use a reward score because a reward is a design choice, while "reached the destination" is not.

Source Plain definition of a proportion. Our choice of the four outcome labels follows what MetaDrive reports at the end of an episode (https://arxiv.org/abs/2109.12674).

3.2 The 95% interval (Wilson)

$$\text{interval} = \frac{\hat{p} + \dfrac{z^2}{2n} \pm z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}, \qquad z = 1.96$$

Symbols $\hat{p}$ and n as above. $z = 1.96$ is the point of the standard normal curve that leaves 2.5% on each side, so that 95% of the probability is in between. The $\pm$ gives the lower and the upper end.

Why With 200 episodes the observed 0.925 is only an estimate of the model's true success rate. The interval says how far the truth can plausibly be from 0.925. We use the

Wilson form because the textbook form $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$ breaks down when $\hat{p}$ is near 0 or 1 (it can give an upper end above 1, or a zero-width interval when $k = 0$, which matters for the image-only model with 0 successes).

Derivation Ask: for which true values p would the observed $\hat{p}$ be an ordinary result? That means $|\hat{p} - p| \le z\sqrt{p(1-p)/n}$. Square both sides: $(\hat{p} - p)^2 \le z^2 p(1-p)/n$. Collect terms: $(1 + z^2/n)\,p^2 - (2\hat{p} + z^2/n)\,p + \hat{p}^2 \le 0$. The two roots of this quadratic in p are exactly the formula above; the interval is everything between the roots.

Source E. B. Wilson (1927), *Journal of the American Statistical Association* 22(158): 209–212 (no arXiv version). Why it is preferred over the textbook form: L. D. Brown, T. T. Cai, A. DasGupta (2001), "Interval estimation for a binomial proportion", *Statistical Science* 16(2): 101–133 (no arXiv version).

3.3 Comparing two models on the same episodes (exact McNemar test)

$$p = \min\Big(1,\ 2\sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)}\Big)$$

Symbols b: number of episodes that model A solved and model B did not. c: the opposite. Episodes that both solved, or both failed, are ignored because they carry no information about which model is better. $\binom{b+c}{i}$: the number of ways to choose i episodes out of $b + c$. $2^{-(b+c)}$: the probability of any one particular sequence of fair coin flips. p: the probability of seeing a split at least this lopsided if the two models were really equally good.

Why Both models drive the *same* 200 seeds, so the results are paired. Comparing two separate intervals throws away that pairing and is too cautious. The paired test uses only the episodes where the models disagree. "Exact" means we use the binomial counts themselves instead of a normal-curve approximation, which is needed when $b + c$ is small (for example 3 against 10).

Derivation If A and B were equally good, each of the $b + c$ disagreeing episodes would favour A or B with probability $\frac{1}{2}$, independently. So the number favouring A follows a binomial distribution with $b + c$ trials and probability $\frac{1}{2}$. The probability of an outcome as extreme as $\min(b, c)$ on the small side is the sum of the binomial terms for $0, \ldots, \min(b, c)$; doubling it makes the test two-sided (a difference in either direction counts); the $\min(1, \cdot)$ keeps $p \le 1$.

Source Q. McNemar (1947), *Psychometrika* 12: 153–157 (no arXiv version). The exact version is the sign test on the disagreeing pairs.

Reading p $p < 0.05$ means the difference is unlikely to be chance. It does *not* say the difference is large, and $p > 0.05$ does not prove the models are equal; it says 200 episodes were not enough to tell.

3.4 Energy

$$\begin{aligned}
W_{\text{inertia}} &= \tfrac{1}{2}\, m\,(v_1^2 - v_0^2) \\
W_{\text{air}} &= k\,\Delta t\,\tfrac{1}{4}\,(v_0^3 + v_0^2 v_1 + v_0 v_1^2 + v_1^3), \quad k = \tfrac{1}{2}\rho C_d A \\
W_{\text{roll}} &= C_{rr}\, m\, g\, \tfrac{v_0+v_1}{2}\,\Delta t \\
W_{\text{step}} &= \max\big(0,\ W_{\text{inertia}} + W_{\text{air}} + W_{\text{roll}}\big), \qquad E_{\text{Wh}} = \frac{\sum W_{\text{step}}}{3600}, \qquad D_{\text{km}} = \frac{1}{1000}\sum \tfrac{v_0+v_1}{2}\Delta t
\end{aligned}$$

Symbols $\Delta t = 0.1\,\mathrm{s}$: one simulator step. v_0, v_1 (m/s): speed at the start and the end of the step. m (kg): car mass, read from the simulator's vehicle. $\rho = 1.225\,\mathrm{kg/m^3}$: air density at sea level. $C_d = 0.23$: drag coefficient (how streamlined the car is). $A = 2.22\,\mathrm{m^2}$: frontal area. $C_{rr} = 0.01$: rolling-resistance coefficient (tyre losses per unit weight). $g = 9.81\,\mathrm{m/s^2}$: gravity. W (joules): mechanical work the wheels must supply in the step. 3600: joules per watt-hour. E_{Wh}: total energy of the episode; D_{km}: distance of the episode.

Why MetaDrive has no battery or fuel model, but the energy question needs a number. Newtonian physics gives one from the only thing every model produces, a speed history. Using a single formula for all five models means that differences in energy come from the way the model drives, not from the formula. **Our choice:** the declared vehicle constants (declared assumptions in the range of a modern electric passenger car, not measured on a real car) and the decision to clip at zero (no regenerative braking).

Derivation The wheels must supply three things. (i) *Change of kinetic energy*: $\frac{1}{2}mv^2$ at the end minus at the start, which gives W_{inertia}. (ii) *Air drag*: the drag force is $\frac{1}{2}\rho C_d A v^2$, so the power is force times speed, $\frac{1}{2}\rho C_d A v^3 = kv^3$, and the work in a step is $k\int v^3\,dt$. Inside a step the speed is assumed to change in a straight line, $v(s) = v_0 + (v_1 - v_0)s$ for $s \in [0,1]$, so $\int_0^1 v(s)^3 ds = \frac{v_1^4 - v_0^4}{4(v_1 - v_0)} = \frac{1}{4}(v_0^3 + v_0^2 v_1 + v_0 v_1^2 + v_1^3)$; multiplying by $k\Delta t$ gives W_{air}. (iii) *Rolling resistance*: the force $C_{rr}mg$ is constant, the power is $C_{rr}mg\,v$, and the average speed over a step with straight-line speed is $\frac{v_0+v_1}{2}$, which gives W_{roll}. Braking gives a negative sum; a car without regeneration cannot recover it, so W_{step} is clipped at 0. Distance is the average speed times Δt, summed.

Why drag matters for the caution-aware model
Drag work grows with v^3, so slowing from 8 to 6 m/s cuts the drag work by a factor $(6/8)^3 \approx 0.42$. This is why a modest speed reduction produces a large energy reduction.

Source Standard longitudinal vehicle dynamics (work–energy theorem, quadratic drag, constant rolling resistance), e.g. L. Guzzella and A. Sciarretta, *Vehicle Propulsion Systems*, Springer (textbook, no arXiv version). The trapezoid-style averaging over a step is our choice.

Limits Flat road, no regeneration, no motor or battery losses, no auxiliaries. The result is mechanical work at the wheels of a declared car and is meaningful only as a comparison between models.

3.5 Detector retention, keep probability and caution

$$r(\kappa,\ell) = \frac{\text{vehicles detected with weather } (\kappa,\ell)}{\text{vehicles detected in clear weather}} \qquad \text{measured once, offlin}$$

$$p_{\mathrm{keep}}(\kappa,\ell) = \text{straight-line interpolation through } (0,1),\ (\ell_1, \min(1,r_1)), \ldots, (\ell_6, \min(1,r_6))$$

$$c = \mathrm{clip}\,(1 - p_{\mathrm{keep}}(\kappa,\ell),\ 0,\ 1)$$

Symbols κ: kind of weather (fog or rain). ℓ: intensity of the teammates' weather filter, from 0 (none) to 1 (strongest). r: retention, the fraction of the vehicles that the detector still finds. $\ell_1, \ldots, \ell_6$: the six intensities that were measured (fog 0.15, 0.3, 0.45, 0.6, 0.8, 1.0; rain 0.1, 0.2, 0.3, 0.45, 0.6, 1.0). p_{keep}: the probability that a vehicle is *kept* in the grid at one simulator step; each vehicle is kept or dropped independently at every step. $\min(1,\cdot)$: retention above 1 is measurement noise (a fog frame gave

more detections than its clear twin), so it is capped. c: caution, between 0 (fully reliable perception) and 1 (almost nothing is seen).

How retention was measured

`bev_rl/adaptive/measure_paired.py`. On 20 scenes (seeds 1000 to 1019) up to three traffic vehicles were placed around a stationary car (at about 8, 10 and 14 m). The *same* camera frames were processed once clean and once with each weather condition (two random weather draws), and the detector counts were compared. Because the frames are paired, the ratio reflects the weather and not different scenes.

Why measure it offline, once

(a) Running the whole camera pipeline (five cameras, detector, stereo depth) costs about 2 s per simulator step, against 0.2 s for the grid; 200 episodes of 800 steps in four weather conditions with five models would take many days of computation. (b) Measuring once turns the pipeline into a single table that the fast tests reuse, and the same table defines how many vehicles disappear in every test. The price is that the grid models are tested with an *ideal* detector plus a measured loss rate, not with the real detector (stated in the honest-limit box above).

Limits of the measurement

Only up to three vehicles at 8 to 14 m with a stationary car, 20 scenes. Distant vehicles are probably missed more than close ones, so the table is likely *optimistic* for far vehicles. Vehicles are dropped independently at each step, so a real detector that misses the same vehicle for many frames would be harsher.

Why $c = 1 - p_{\text{keep}}$

Our choice. It is the simplest number that is 0 when perception is complete, rises as vehicles are lost, and needs no extra measurement. Any monotone function of reliability would be a valid alternative.

What the caution-aware model is told and what it is not

You are right that the model is *not* told "this is fog" or "this is rain". It receives a single number, c, that says how cautious to be. Two honest qualifications. (1) In our tests c is looked up from the known weather condition through the table above, so it is the weather label in disguise: it carries exactly the information "fog 1.0 gives 0.73, rain 1.0 gives 0.96". (2) The same table also decides how many vehicles are removed from the grid, so the number the model receives is perfectly consistent with what it sees. A deployed system would have to *estimate* c from the camera, for example from detector confidence or an image-quality score, and its estimate would be noisy. Whether the learned behaviour survives that noise is untested.

Is weather detection a solved problem? Partly. Classifying a driving image as clear, fog, rain or snow with a neural network is well studied and reaches high accuracy on easy datasets. The authors of a driving-specific dataset report that the same classifier does much worse on their harder images, and that the task is not saturated (Dhananjaya, Ravi Kumar, Yogamani, https://arxiv.org/abs/2104.14042). Estimating *how much* the weather reduces visibility, rather than naming the weather, is a separate and less settled task (for example pixel-wise visibility under fog, https://arxiv.org/abs/2112.04278; overview of perception under adverse weather, https://arxiv.org/abs/2112.08936). So a trained weather or visibility estimator is a reasonable replacement for the lookup table, and building on this literature is the natural next step. It has not been done here.

3.6 The teacher and its dependence on caution

$$v_{\text{target}}(c) = 30\Big(1 - \frac{c}{3}\Big)\ \text{km/h}, \qquad s_{\text{gap}}(c) = 10 + 15\,c\ \text{m}$$

Symbols c caution (0 to 1). v_{target}: the speed the rule-based teacher tries to hold when the road is free. s_{gap}: the distance it tries to keep to the vehicle ahead.

Why A cautious driver drives slower and keeps more room. The linear mapping gives the two end points that were chosen: 30 km/h and 10 m at $c = 0$ (MetaDrive's stock driver), 20 km/h and 25 m at $c = 1$. **Our choice:** the end points and the linear form; there is no data behind them. The student copies these labels and so learns the dependence.

Source of the teacher

The teacher is MetaDrive's rule-based driver in the family of the Intelligent Driver Model, which sets the acceleration from the current speed, the desired speed and the gap to the vehicle ahead: M. Treiber, A. Hennecke, D. Helbing, https://arxiv.org/abs/cond-mat/0002177. The mapping above changes its two parameters (desired speed, desired gap).

3.7 Imitation training: behaviour cloning and dataset aggregation

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \big\|\pi_\theta(o_i) - a_i^{\text{teacher}}\big\|^2, \qquad \pi_{\text{run},t} = \begin{cases} \text{teacher action} & \text{with probability } \beta_t = 0.5^{\,t} \\ \pi_\theta\text{'s action} & \text{otherwise} \end{cases}$$

Symbols π_θ: the driving network with adjustable numbers θ; o_i: what it sees in sample i (grid and vehicle-state numbers); a_i^{teacher}: the two numbers (steering, throttle or brake) that the teacher would have chosen in that state; N: number of samples collected so far; $\|\cdot\|^2$: sum of squared differences of the two numbers. t: round of dataset aggregation (0, 1, 2, ...); β_t: how often the teacher, rather than the network, is executed while collecting data in round t (round 0: 1, so the teacher drives).

Why *Behaviour cloning* (squared error on recorded actions) is supervised learning and trains in minutes. Its weakness is that the network only sees states the teacher visited; after one small mistake it is in a state that is not in the data, and errors grow. *Dataset aggregation* fixes this: the network drives (partly), the teacher labels the states the network actually reached, and those labels are added and the network is retrained on everything. The share of teacher control β_t shrinks each round so that the data increasingly come from the network's own mistakes.

Derivation of the idea

If the network errs with probability ϵ per step on the teacher's states, behaviour cloning can accumulate error of the order ϵT^2 over T steps, because each mistake moves the car to states the teacher never showed. Training on the network's own state distribution, as dataset aggregation does, reduces this to the order ϵT (Ross, Gordon and Bagnell prove this through a reduction to no-regret online learning).

Source S. Ross, G. Gordon, J. A. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning" (DAgger), https://arxiv.org/abs/1011.0686. The schedule $\beta_t = 0.5^t$ and the noise level are our settings. The optimiser, Adam: D. Kingma and J. Ba, https://arxiv.org/abs/1412.6980.

Network output

Both outputs pass through tanh, so steering and throttle are between -1 and 1.

3.8 Reinforcement learning: the objective and the reward terms

$$L^{\text{clip}}(\theta) = \mathbb{E}_t\Big[\min\left(\rho_t(\theta)\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t\right)\Big], \qquad \rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

Symbols s_t, a_t: state and action at step t. $\pi_\theta(a \mid s)$: the probability that the current policy takes action a in state s; θ_{old}: the policy that collected the data. ρ_t: how much more (or less) likely the new policy makes the action that was taken. $\hat{A}_t$: advantage, an estimate of how much better than average that action was. ε: the clip range (0.1 in attempt 2; the other attempts' values are in the training records). $\mathbb{E}_t$: average over the collected steps.

Why A policy gradient step that is too large can wreck a working policy. Clipping the ratio ρ_t removes the incentive to move the policy further than $1 \pm \varepsilon$ from the one that collected the data, which allows several update epochs on the same batch.

Source J. Schulman et al., "Proximal policy optimization algorithms", https://arxiv.org/abs/1707.06347.

Table 3: The terms of the two reward formulas in the reinforcement-learning section. Every term is our design choice; the form (progress minus penalties plus terminal reward) is the usual shaping style in driving simulators such as MetaDrive.

Term	Meaning of the symbols	Why it is there
$\min(\Delta s, v_{cap}\Delta t)/(v_{ref}\Delta t)$	Δs: metres driven in the step; v_{cap}: speed cap; $v_{ref} = 10$ m/s: reference speed; $\Delta t = 0.1$ s.	Reward progress, capped so that speeding is not paid.
$-0.5\,d^2$	d: sideways distance from the lane centre (m).	Stay in the lane; squared so large errors cost much more.
$-5\,\psi^2$	ψ: angle between car and lane direction (rad).	Keep the car pointing along the road.
$-0.1\lVert a_t - a_{t-1}\rVert^2$	a_t: action (steering, throttle or brake).	Smooth control, no twitching.
$-0.002\,j^2$	j: jerk, rate of change of acceleration.	Passenger comfort and smoother speed.
$-0.2\max(0, -a_t^{thr})$ (version 1)	a_t^{thr}: throttle or brake command, negative = braking.	Mild penalty on braking (removed in version 3).
$-0.5\big((v - v_{cap})_+/2\big)^2$ (version 1)	$(x)_+ = \max(0, x)$; v: speed.	Penalty for exceeding the speed cap.
$-2(1 - \tau/3)_+^2$ (version 3)	τ: predicted time to collision (s).	Rises as a collision gets closer than 3 s.
$-(1 - g/6)_+^2$ (version 3)	g: gap to the vehicle ahead (m).	Penalty for following closer than 6 m.
$-0.02\,E_t$ (version 3)	E_t: energy of the step in Wh (Section 3, energy).	Make energy cost part of the objective.
$-0.2[v < 0.5]$	$[\cdot]$ is 1 if true, 0 otherwise.	Stops the policy from learning to stand still.
R_{end}	Terminal reward: $+20/-50$ (version 1), $+50/-100$ (version 3).	Large reward for reaching the goal, large penalty for crashing or leaving the road.

3.9 Grid geometry

The bird's-eye-view grid has 64×64 cells over 20 m × 20 m, so one cell is $20/64 = 0.3125$ m. The window starts 2.5 m behind the car and reaches 17.5 m ahead, which is why vehicles further than 17.5 m are invisible to every grid model. This is a design of the teammates' code (`bev_state/`), not ours.

3.10 References for this section

- Ross, Gordon, Bagnell (DAgger): https://arxiv.org/abs/1011.0686
- Treiber, Hennecke, Helbing (Intelligent Driver Model): https://arxiv.org/abs/cond-mat/0002177
- Schulman et al. (proximal policy optimisation): https://arxiv.org/abs/1707.06347
- Kingma, Ba (Adam): https://arxiv.org/abs/1412.6980
- Li et al. (MetaDrive): https://arxiv.org/abs/2109.12674
- Dhananjaya, Ravi Kumar, Yogamani (weather and light classification): https://arxiv.org/abs/2104.14042
- Visibility estimation under fog (DMRVisNet): https://arxiv.org/abs/2112.04278
- Survey of perception under adverse weather: https://arxiv.org/abs/2112.08936
- Without arXiv version: Wilson (1927); Brown, Cai, DasGupta (2001); McNemar (1947); Guzzella and Sciarretta, *Vehicle Propulsion Systems*.

4 Results of all five models together

Table 4: All metrics, five models, four weather conditions, 200 episodes each (seeds 5000–5199, sparse traffic, 800-step limit). Collision, left road and timeout are shares of the episodes. Energy is the estimate of Section 2.4. “not yet evaluated” marks a run that had not finished when this document was built.

Weather	Model	Success	95% interval	Collision	Left road	Timeout	Speed (m/s)	Route completion	Energy (Wh per km)
Clear	Baseline Driving Model Trained Without Traffic	0.850	[0.794, 0.893]	0.125	0.025	0.000	8.09	0.918	85.9
	Baseline Driving Model Trained With Traffic	0.745	[0.680, 0.800]	0.120	0.135	0.000	8.07	0.883	90.7
	Proposed Bird’s-Eye-View Driving Model	0.925	[0.880, 0.954]	0.060	0.010	0.005	7.87	0.939	96.2
	Caution-Study Caution-Aware Driving Model	0.890	[0.839, 0.926]	0.090	0.020	0.000	7.97	0.920	94.1
	Image-Only Driving Model (front camera)	0.130	[0.090, 0.184]	0.200	0.605	0.065	9.20	0.550	276.6
Fog (intensity 1.0)	Baseline Driving Model Trained Without Traffic	0.860	[0.805, 0.901]	0.115	0.025	0.000	8.09	0.920	85.9
	Baseline Driving Model Trained With Traffic	0.750	[0.686, 0.805]	0.120	0.130	0.000	8.07	0.886	90.5
	Proposed Bird’s-Eye-View Driving Model	0.895	[0.845, 0.930]	0.085	0.020	0.000	7.92	0.933	96.5
	Caution-Study Caution-Aware Driving Model	0.895	[0.845, 0.930]	0.050	0.020	0.035	6.15	0.945	62.7
	Image-Only Driving Model (front camera)	0.000	[0.000, 0.019]	0.000	1.000	0.000	5.39	0.110	392.3
Rain (intensity 0.6)	Baseline Driving Model Trained Without Traffic	0.850	[0.794, 0.893]	0.125	0.025	0.000	8.09	0.918	86.0
	Baseline Driving Model Trained With Traffic	0.755	[0.691, 0.809]	0.120	0.125	0.000	8.08	0.886	90.5
	Proposed Bird’s-Eye-View Driving Model	0.900	[0.851, 0.934]	0.085	0.015	0.000	7.94	0.934	94.9
	Caution-Study Caution-Aware Driving Model	0.890	[0.839, 0.926]	0.030	0.020	0.060	5.87	0.951	60.4
	Image-Only Driving Model (front camera)	0.030	[0.014, 0.064]	0.120	0.850	0.000	8.76	0.339	341.1
Rain (intensity 1.0)	Baseline Driving Model Trained Without Traffic	0.850	[0.794, 0.893]	0.125	0.025	0.000	8.09	0.918	85.9
	Baseline Driving Model Trained With Traffic	0.755	[0.691, 0.809]	0.120	0.125	0.000	8.08	0.886	90.5
	Proposed Bird’s-Eye-View Driving Model	0.900	[0.851, 0.934]	0.095	0.005	0.000	7.95	0.932	93.0
	Caution-Study Caution-Aware Driving Model	0.820	[0.761, 0.867]	0.035	0.040	0.105	5.56	0.935	59.7
	Image-Only Driving Model (front camera)	0.010	[0.003, 0.036]	0.020	0.970	0.000	5.60	0.178	320.8

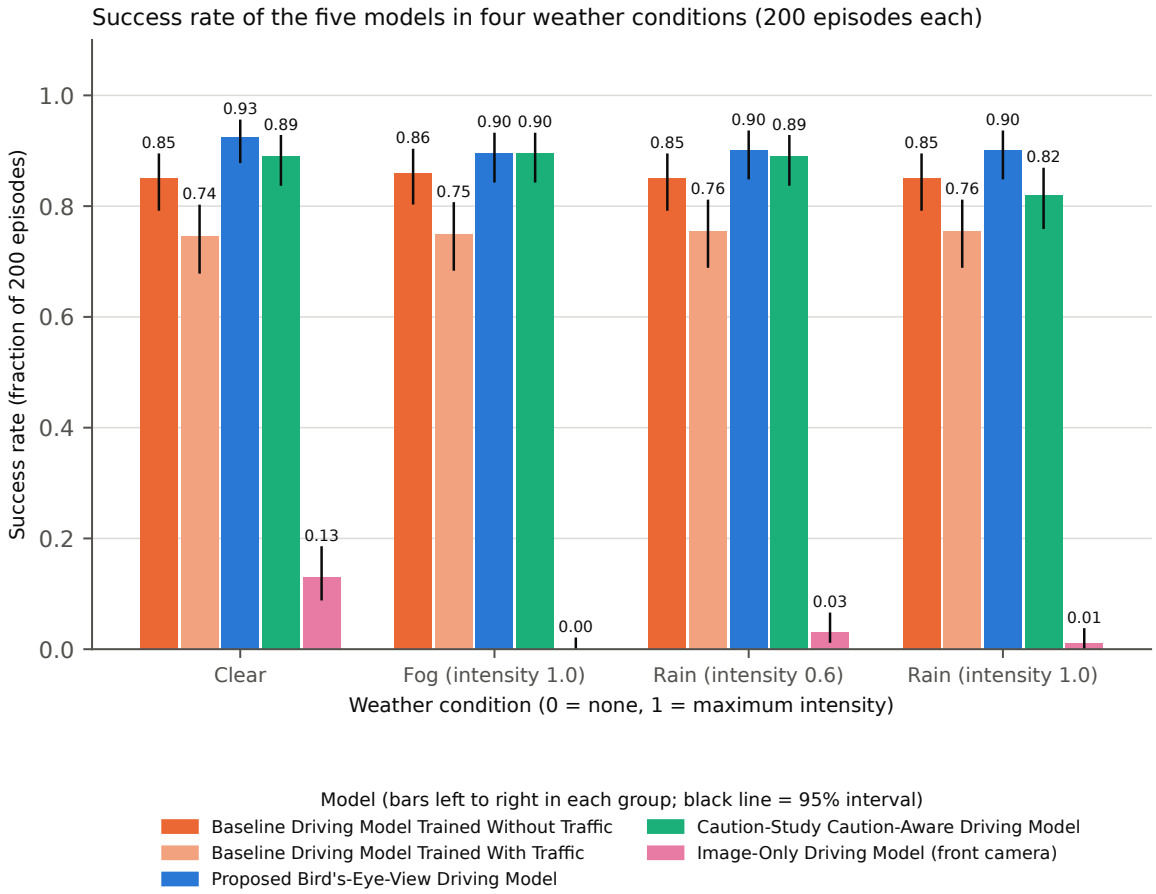


Figure 2: Success rate of the five models in four weather conditions. Each bar is the share of 200 episodes that reached the destination; the black line is the 95% interval. Weather becomes stronger from left to right.

Plain-English reading. The proposed model and the caution-aware model stay between 0.82 and 0.93; the two baselines stay near 0.85 and 0.75; the image-only model is far below all of them. Weather barely moves any of the four grid-based models in the tests that apply it as a measured loss of vehicles; the one clear drop is the caution-aware model in rain 1.0 at the 80-second limit, which is caused by timeouts (Section 7.7).

Research implication. The bird's-eye-view grid is what makes driving possible at this level; the training method (repeated correction) separates the proposed model from the baselines. The weather result is limited to loss of detected vehicles in sparse traffic.

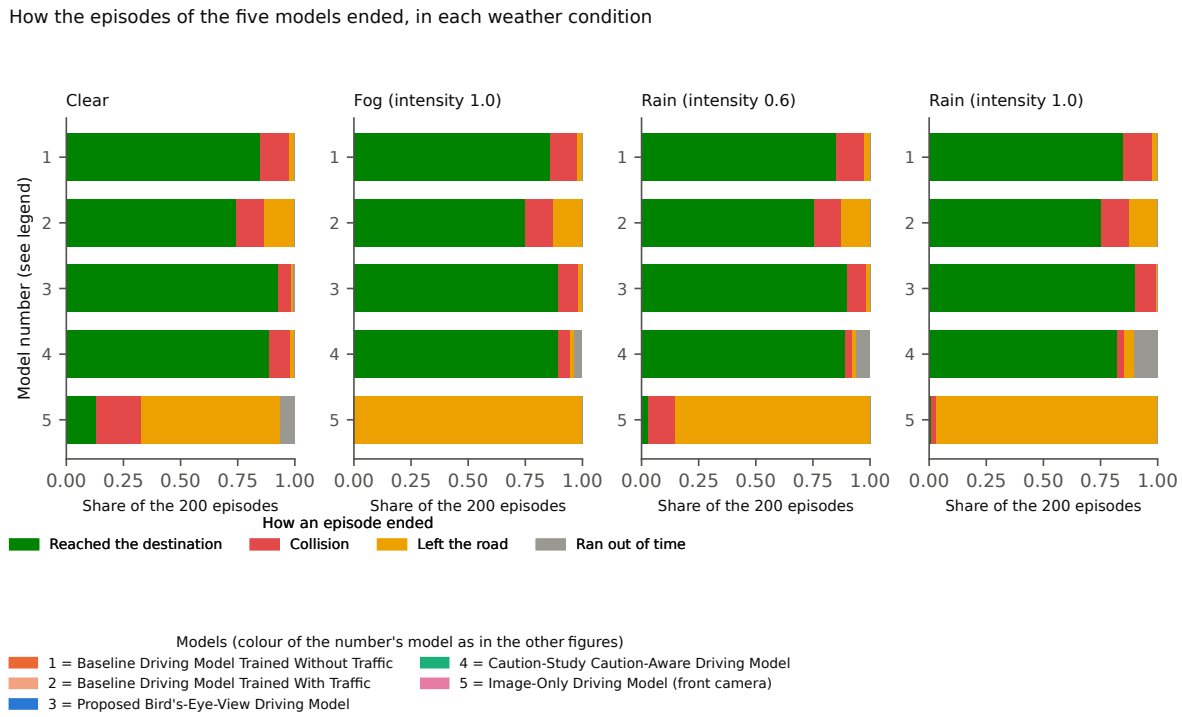


Figure 3: How the episodes ended, per model and weather. Each horizontal bar is one model in one weather condition; its coloured parts add up to the 200 episodes: green = destination reached, red = collision, yellow = left the road, grey = time limit reached.

Plain-English reading. The baselines fail mainly by collisions (and, for the one trained with traffic, by leaving the road); the proposed model fails rarely and mostly by collision; the caution-aware model replaces collisions by timeouts as the weather gets stronger; the image-only model mostly leaves the road.

Research implication. Models fail in different ways, so success alone hides what to fix. Leaving the road points at steering and road understanding; collisions point at seeing and reacting to other vehicles; timeouts point at speed choice.

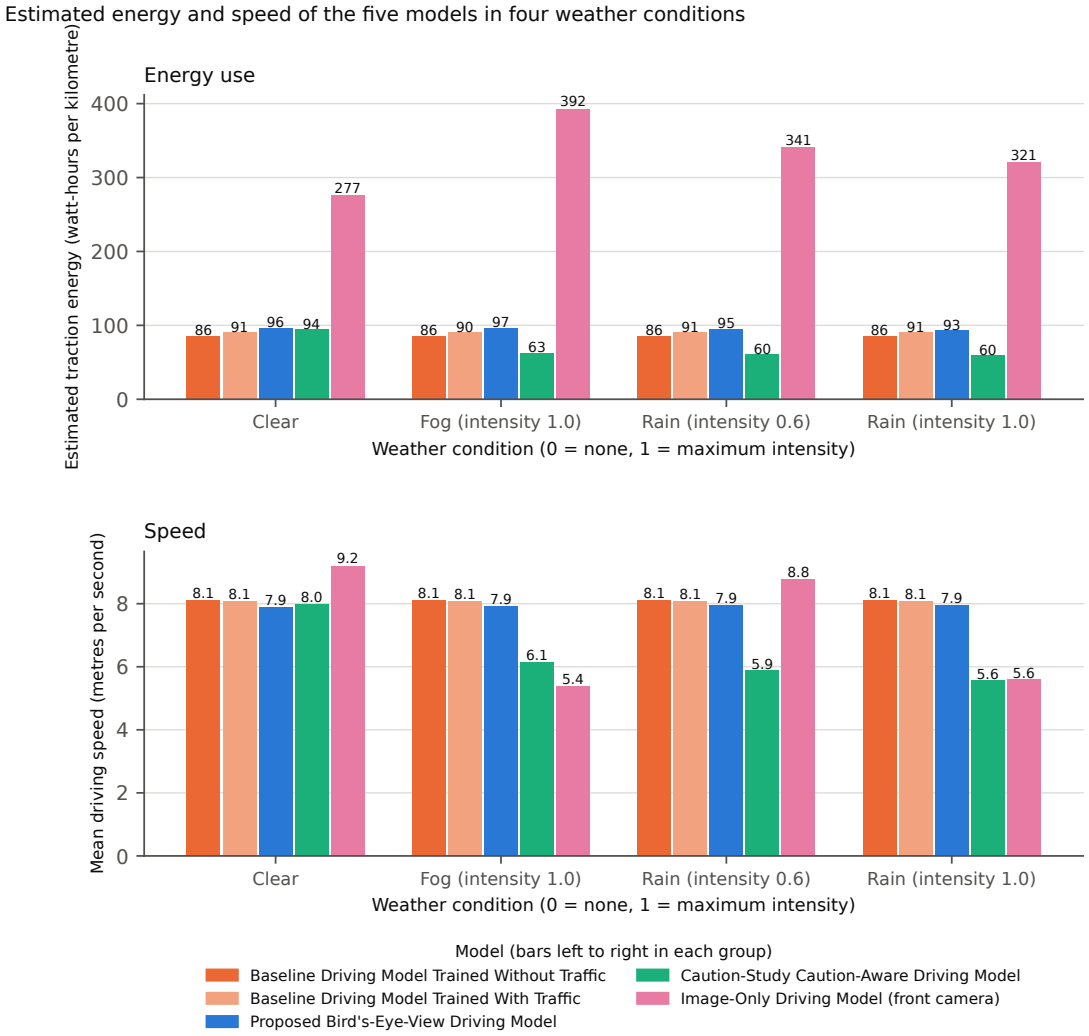


Figure 4: Estimated energy and mean speed of the five models. Top: estimated traction energy in watt-hours per kilometre. Bottom: mean speed in metres per second. Same four weather conditions.

Plain-English reading. Baselines and the proposed model use about 86 to 96 watt-hours per kilometre at about 8 metres per second. The caution-aware model drops to about 60 in fog and rain by driving at 5.6 to 6.2 metres per second. The image-only model is the fastest and the most energy-hungry.

Research implication. Lower energy came only from lower speed. A speed-matched comparison would be needed to say that the caution-aware model is more *efficient* and not simply slower.

Table 5: Paired comparison of every model against the image-only model on the same episodes. Positive differences favour the model named in the second column.

Weather	Model	Difference in success (model minus image-only)	Only this model succeeded	Only image-only succeeded	Exact McNemar p
Clear	Baseline Driving Model Trained Without Traffic	+0.720	144	0	< 0.0001
	Baseline Driving Model Trained With Traffic	+0.615	128	5	< 0.0001
	Proposed Bird's-Eye-View Driving Model	+0.795	159	0	< 0.0001
	Caution-Study Caution-Aware Driving Model	+0.760	152	0	< 0.0001
Fog (intensity 1.0)	Baseline Driving Model Trained Without Traffic	+0.860	172	0	< 0.0001
	Baseline Driving Model Trained With Traffic	+0.750	150	0	< 0.0001
	Proposed Bird's-Eye-View Driving Model	+0.895	179	0	< 0.0001
	Caution-Study Caution-Aware Driving Model	+0.895	179	0	< 0.0001
Rain (intensity 0.6)	Baseline Driving Model Trained Without Traffic	+0.820	164	0	< 0.0001
	Baseline Driving Model Trained With Traffic	+0.725	148	3	< 0.0001
	Proposed Bird's-Eye-View Driving Model	+0.870	174	0	< 0.0001
	Caution-Study Caution-Aware Driving Model	+0.860	172	0	< 0.0001
Rain (intensity 1.0)	Baseline Driving Model Trained Without Traffic	+0.840	168	0	< 0.0001
	Baseline Driving Model Trained With Traffic	+0.745	150	1	< 0.0001
	Proposed Bird's-Eye-View Driving Model	+0.890	178	0	< 0.0001
	Caution-Study Caution-Aware Driving Model	+0.810	162	0	< 0.0001

Table 6: Paired comparison among the four grid-based models in clear weather (200 episodes). From the technical report.

Comparison	Difference in success	Only first succeeded	Only second succeeded	Exact McNemar p
Proposed against Baseline without traffic	+0.075	—	—	0.0007
Proposed against Baseline with traffic	+0.180	—	—	< 0.0001
Caution-aware against Proposed	−0.035	3	10	0.092
Caution-aware against Baseline without traffic	+0.040	11	3	0.057
Caution-aware against Baseline with traffic	+0.145	32	3	< 0.0001

5 Model 1 and model 2: the two baseline driving models (teammates)

5.1 What they are

Two copies of the same design, trained from five demonstrations each. Model 1 was trained on drives without other vehicles; model 2 on drives with sparse traffic. Both are on GitHub (`bev_policy/checkpoints/`).

5.2 What they see and the network

A 64×64 bird's-eye-view grid (20 m wide and long: 17.5 m ahead, 2.5 m behind; cell 0.3125 m) with values for unknown, free, own car and detected vehicle, *no road drawing*, and six vehicle-state numbers (speed, route progress, sideways offset from the lane centre, heading error, curvature ahead, distance to goal). The network is the shared `BEVScalarPolicy` (110 530 adjustable numbers): three convolution layers (32, 64, 96 maps) on the grid, a small network (6→64→32) on the numbers, and a head (160→128→2) with output steering and throttle or brake, each between −1 and 1.

5.3 How they were trained

Behaviour cloning: MetaDrive's stock driving rule (Intelligent Driver Model) drives five episodes, only episodes that reach the destination are kept, and the network is fitted to the recorded actions (mean squared error, at most 60 epochs, batch 64, learning rate $3 \cdot 10^{-4}$, early stopping after 10 epochs without improvement). **Weakness:** the network sees only places the expert visited, so after a small error it has no example of how to recover.

5.4 How they were tested

As in Section 2. The weather proxy removes vehicles from the grid; the models are not given any weather input.

5.5 Results

Success 0.850 (without traffic) and 0.745 (with traffic) in clear weather, with 0.85–0.86 and 0.75–0.755 in the three weather conditions (Table 4). The one trained *with* traffic is the *worse* one: it leaves the road in 13.5% of clear episodes (the other 2.5%), collisions about 12% in both. Energy 85.9 and 90.7 watt-hours per kilometre at 8.1 m/s.

5.6 What we can and cannot say

We can say	Because
Training with traffic did not help these baselines; it was 0.105 worse in clear weather.	200 paired episodes, 0.745 against 0.850.
Both baselines are flat across the four weather conditions.	Success within 0.015 in each, in the vehicle-loss proxy.
We cannot say	**Because**
That 0.745 contradicts the README value 0.90.	The README reports 9 of 10 episodes (interval 0.596–0.982), which includes 0.745.
That the baselines are robust to real fog and rain in the camera chain.	The perception damage is measured as vehicle loss only.

6 Model 3: the Proposed Bird's-Eye-View Driving Model

6.1 What it is

The same network family as the baselines, with two changes to what it *sees* and a different way of *teaching*.

6.2 What it sees and the network

- The grid has the **road drawn in**: the route centre line (value 0.5) and the road boundaries (0.8), taken from the simulator's map in the car's frame.
- It gets **three** vehicle-state numbers (speed, route progress, distance to goal). Sideways offset, heading error and curvature are *not* given as numbers, because the drawn road already shows them.

Network: the same `BEVScalarPolicy`, 110 530 adjustable numbers.

6.3 How it was trained

Dataset aggregation: the model drives, a rule-based teacher says what it should have done at each state the model reached, and these labels are added to the training data. Six rounds (0 to 5).

Round 0: the teacher drives. Later rounds: the teacher's action is executed with probability 0.5^t, otherwise the model's. Each round: 8 episodes at each of five traffic densities (0, 0.05, 0.1, 0.2, 0.3) on training seeds 2000–2999, Gaussian action noise 0.1, retraining on all data so far (Adam, learning rate 10^{-3}, batch 128, mean squared error; 24 epochs in round 0, 8 later). The saved model is round 5, trained on 73 032 labelled steps. Teacher: the stock Intelligent Driver Model on the exact simulator state.

6.4 How it was tested

As in Section 2; no weather input; weather acts through vehicle loss only.

6.5 Results

Clear 0.925 (interval 0.880–0.954), fog 1.0 0.895, rain 0.6 0.900, rain 1.0 0.900. It is better than the baselines in clear weather by +0.075 ($p = 0.0007$) and +0.180 ($p < 0.0001$). 15 of 200 clear episodes fail; a collision-aware teacher solves 14 of those 15, which points to information that the model's grid does not contain (late, coarse view of other vehicles). Energy 93.0–96.5 watt-hours per kilometre at 7.9 m/s.

6.6 What we can and cannot say

We can say	Because
Repeated drive-and-correct training with the road drawn in beats five-demonstration cloning.	+0.075 and +0.180 on 200 paired episodes.
Loss of detected vehicles at the measured rates did not lower its success in sparse traffic.	0.895–0.925 in all four conditions; the pre-registered test of harm failed.
We cannot say	**Because**
Which of the two changes (road drawing, training method) caused the gain.	They were not separated at 200 episodes.
That it is better than its own teacher.	The collision-aware teacher scores 0.950; the difference is not significant.
That it works with a real detector and camera.	Tests used a measured loss of vehicles on top of the simulator's vehicle list.

7 Model 4: the Caution-Study Caution-Aware Driving Model

7.1 What it is

The proposed design with **one extra number**, "caution" between 0 and 1. Caution is derived from the measured detector reliability: $c = 1 -$ retention (0 in clear weather).

7.2 What it sees and the network

Same grid and three vehicle-state numbers as model 3, plus caution through a small extra branch (1→16) joined before the head (176→128→2); 112 610 adjustable numbers. *The caution value is given to the model from the known weather; the model does not sense it.*

7.3 How it was trained

Dataset aggregation as for model 3, but the **teacher also depends on caution**: at caution 0 it drives at 30 km/h with a 10 m gap; at caution 1 at 20 km/h with a 25 m gap. Five rounds of 40 episodes, 6 epochs per round, action noise 0.1, caution per episode half uniform in $[0, 1]$ and half derived from weather. The saved model is the round with the best validation result, *round 1* (35 503 labelled steps). A control with the caution input fixed at 0 was trained the same way.

7.4 How it was tested

As in Section 2, with caution computed from the weather condition (clear: 0).

7.5 Results

Success 0.890, 0.895, 0.890, 0.820 (clear, fog 1.0, rain 0.6, rain 1.0). In weather it slows to 5.6–6.2 m/s and its estimated energy falls to 59.7–62.7 watt-hours per kilometre, about 35% lower than the proposed model (93.0–96.5). Collisions fall to 3–5%, but timeouts rise to 3.5%, 6% and 10.5%.

7.6 Why it differs from the proposed model in clear weather

It is a different network, trained in a separate run on different labels, so it behaves differently even at caution 0. It solves 178 of 200 clear episodes, the proposed model 185; they disagree on 13 episodes (3 vs 10, $p = 0.092$). The control trained the same way with caution fixed at 0 scores 0.895, the same as model 4, so run-to-run differences of a few points exist at 200 episodes.

7.7 Longer time limit (post-hoc check)

Every extra failure of model 4 in weather was a timeout, so it was re-run in the three weather conditions with a 160-second limit (1600 steps; same seeds), and then all five models were put on the same 160-second footing (Table 7). A longer limit can only change an episode that ran out of time at 80 seconds, so for the other models only those episodes were re-run: one episode of the proposed model (clear, seed 5151, now a success) and 13 of the image-only model (clear). The two baselines, and the proposed and caution-aware models in fog and rain, had no timeouts to re-run. The limit was changed *after* seeing the timeouts, so the 160-second table is post-hoc.

Table 7: All five models with the 160-second limit (1600 steps). 200 episodes each, seeds 5000 to 5199, sparse traffic. Eleven image-only episodes in clear weather still time out at 160 seconds; there are no other timeouts.

Weather	Model	Success	95% interval	Collision	Left road	Timeout	Speed (m/s)	Energy (Wh per km)
Clear	Baseline Driving Model Trained Without Traffic	0.850	[0.794, 0.893]	0.125	0.025	0.000	8.09	85.9
	Baseline Driving Model Trained With Traffic	0.745	[0.680, 0.800]	0.120	0.135	0.000	8.07	90.7
	Proposed Bird's-Eye-View Driving Model	0.930	[0.886, 0.958]	0.060	0.010	0.000	7.89	96.0
	Caution-Study Caution-Aware Driving Model	0.890	[0.839, 0.926]	0.090	0.020	0.000	7.97	94.1
	Image-Only Driving Model (front camera)	0.130	[0.090, 0.184]	0.210	0.605	0.055	9.20	276.2
Fog (intensity 1.0)	Baseline Driving Model Trained Without Traffic	0.860	[0.805, 0.901]	0.115	0.025	0.000	8.09	85.9
	Baseline Driving Model Trained With Traffic	0.750	[0.686, 0.805]	0.120	0.130	0.000	8.07	90.5
	Proposed Bird's-Eye-View Driving Model	0.895	[0.845, 0.930]	0.085	0.020	0.000	7.92	96.5
	Caution-Study Caution-Aware Driving Model	0.940	[0.898, 0.965]	0.050	0.010	0.000	6.15	62.5
	Image-Only Driving Model (front camera)	0.000	[0.000, 0.019]	0.000	1.000	0.000	5.39	392.3
Rain (intensity 0.6)	Baseline Driving Model Trained Without Traffic	0.850	[0.794, 0.893]	0.125	0.025	0.000	8.09	86.0
	Baseline Driving Model Trained With Traffic	0.755	[0.691, 0.809]	0.120	0.125	0.000	8.08	90.5
	Proposed Bird's-Eye-View Driving Model	0.900	[0.851, 0.934]	0.085	0.015	0.000	7.94	94.9
	Caution-Study Caution-Aware Driving Model	0.955	[0.917, 0.976]	0.025	0.020	0.000	5.87	60.2
	Image-Only Driving Model (front camera)	0.030	[0.014, 0.064]	0.120	0.850	0.000	8.76	341.1
Rain (intensity 1.0)	Baseline Driving Model Trained Without Traffic	0.850	[0.794, 0.893]	0.125	0.025	0.000	8.09	85.9
	Baseline Driving Model Trained With Traffic	0.755	[0.691, 0.809]	0.120	0.125	0.000	8.08	90.5
	Proposed Bird's-Eye-View Driving Model	0.900	[0.851, 0.934]	0.095	0.005	0.000	7.95	93.0
	Caution-Study Caution-Aware Driving Model	0.935	[0.892, 0.962]	0.035	0.030	0.000	5.56	59.7
	Image-Only Driving Model (front camera)	0.010	[0.003, 0.036]	0.020	0.970	0.000	5.60	320.8

Table 8: Caution-aware model at 80 and 160 seconds, next to the proposed model at 80 seconds. 200 episodes each.

Weather	Model	Step limit	Success	Collision	Left road	Timeout	Speed (m/s)	Energy (Wh per km)
Fog (intensity 1.0)	Proposed Bird's-Eye-View Driving Model	800 (80 s)	0.895	0.085	0.020	0.000	7.92	96.5
	Caution-Study Caution-Aware Driving Model	800 (80 s)	0.895	0.050	0.020	0.035	6.15	62.7
	Caution-Study Caution-Aware Driving Model	1600 (160 s)	0.940	0.050	0.010	0.000	6.15	62.5
Rain (intensity 0.6)	Proposed Bird's-Eye-View Driving Model	800 (80 s)	0.900	0.085	0.015	0.000	7.94	94.9
	Caution-Study Caution-Aware Driving Model	800 (80 s)	0.890	0.030	0.020	0.060	5.87	60.4
	Caution-Study Caution-Aware Driving Model	1600 (160 s)	0.955	0.025	0.020	0.000	5.87	60.2
Rain (intensity 1.0)	Proposed Bird's-Eye-View Driving Model	800 (80 s)	0.900	0.095	0.005	0.000	7.95	93.0
	Caution-Study Caution-Aware Driving Model	800 (80 s)	0.820	0.035	0.040	0.105	5.56	59.7
	Caution-Study Caution-Aware Driving Model	1600 (160 s)	0.935	0.035	0.030	0.000	5.56	59.7

Table 9: Paired tests at 160 seconds. First rows: caution-aware against proposed. Second rows: a control run in which the caution input is forced to 1 in every weather, including clear, against the caution-aware model (post-hoc control, results in the folder sparse_always_cautious).

Weather	Comparison	Difference in success	Only first succeeded	Only second succeeded	Exact McNemar p
Clear	caution-aware minus proposed	-0.040	3	11	0.057
	always-cautious control minus caution-aware	+0.040	21	13	0.229
Fog (intensity 1.0)	caution-aware minus proposed	+0.045	19	10	0.136
	always-cautious control minus caution-aware	-0.005	7	8	1.000
Rain (intensity 0.6)	caution-aware minus proposed	+0.055	20	9	0.061
	always-cautious control minus caution-aware	-0.020	3	7	0.344
Rain (intensity 1.0)	caution-aware minus proposed	+0.035	20	13	0.296
	always-cautious control minus caution-aware	+0.010	5	3	0.727

At 160 seconds all timeouts of model 4 disappear: 0.940, 0.955 and 0.935 in fog 1.0, rain 0.6 and rain 1.0, against 0.895, 0.900 and 0.900 for the proposed model, a gain of 3.5 to 5.5 points in every bad-weather condition. The gain is in the expected direction in all three, but no single comparison is significant ($p = 0.136$, 0.061, 0.296), and in clear weather the caution-aware model is 4 points lower ($p = 0.057$). Its collision rate in weather (2.5 to 5.0%) is below the proposed model's (8.5 to 9.5%).

Where does the gain come from? A control. When caution is forced to 1 in every weather, *including clear*, the same network scores 0.930, 0.935, 0.935 and 0.945 (clear, fog 1.0, rain 0.6, rain 1.0) at 160 seconds, with the same speed (5.5 m/s) and energy (about 59 watt-hours per kilometre)

in every condition; none of its differences from the caution-aware model is significant (second rows of Table 9). So in sparse traffic the improvement is explained by *driving more cautiously*, which the weather-derived caution value switches on; it is not evidence that the model reads the weather better. The adaptive part is that caution is switched on only when perception is degraded, so clear-weather driving keeps its normal speed.

7.8 What we can and cannot say

We can say	Because
The model shows *learned adaptive behaviour*: it slows down by itself when the caution value is high.	The control with caution fixed at 0 does not slow down; speed fell from about 8.0 to 5.6–6.2 m/s.
In fog and rain it used about 35% less estimated energy, and with a 160-second limit its success was 3.5 to 5.5 points *above* the proposed model's in all three conditions.	Tables 7 and 8; each difference alone is not significant ($p = 0.061$ to 0.296).
Its collision rate in weather is lower (2.5 to 5.0% against 8.5 to 9.5%).	Table 7.
At the original 80-second limit it was significantly lower in rain 1.0 (0.820 against 0.900, $p = 0.033$) because of timeouts.	Table 4.
We cannot say	**Because**
That the gain is specific to weather.	Forcing caution to 1 in every weather, clear included, gives 0.930 to 0.945 at 160 seconds, as high as the caution-aware model; the gain comes from slower driving.
That weather hurt the proposed model in sparse traffic.	It scored 0.895 to 0.930 in all four conditions; the pre-registered test of harm failed.
That it is more energy-*efficient*.	It is slower; no speed-matched control exists.
That the model senses the weather.	The caution value is supplied.
That the 160-second result is pre-registered.	The limit was changed after the timeouts were seen.

8 Model 5: the Image-Only Driving Model (front camera)

8.1 What it is

A model that gets only the front-left camera picture and nothing else: no grid, no road drawing, no speed, no weather label. It is the weak control: "what happens if we skip the bird's-eye-view grid."

8.2 What it sees and the network

The camera picture is rendered at 320×240, resized to 128×96 and scaled to the range 0 to 1. Five convolution layers with batch normalisation (24, 36, 48, 64, 64 maps; kernels 5, 5, 3, 3, 3; all stride 2; output 64×3×4), then a head (768→128, dropout 0.2, 128→2, tanh). It is a different, smaller network type than models 1 to 4.

8.3 How it was trained

Behaviour cloning of the collision-aware teacher on 40 clean-weather episodes (teacher drives with action noise 0, 0.1 and 0.25, so that recoveries are in the data), followed by three dataset-aggregation rounds of 20 episodes (teacher mix 0.30, 0.20, 0.10), 37 692 labelled pictures in total; Adam, learning rate 10^{-3} then $5 \cdot 10^{-4}$, 8 epochs per fit, mild brightness jitter. Training seeds 100–199. It was trained only on clean pictures.

8.4 How it was tested

The same 200 seeds and metrics. In weather, the teammates' fog or rain filter is applied to the picture it receives (a different seed stream per episode), at the same intensities as for the other models.

8.5 Results

Clear: success 0.130 (interval 0.090 to 0.184); 60.5% of episodes end by leaving the road, 20.0% by collision, 6.5% by timeout; speed 9.20 m/s, route completion 0.55, energy 276.6 watt-hours per kilometre. In weather (the same weather filter on the camera picture): fog (intensity 1.0): success 0.000, left the road 1.000, collision 0.000, timeout 0.000, route completion 0.11, speed 5.39 m/s, energy 392.3 Wh per km; rain (intensity 0.6): success 0.030, left the road 0.850, collision 0.120, timeout 0.000, route completion 0.34, speed 8.76 m/s, energy 341.1 Wh per km; rain (intensity 1.0): success 0.010, left the road 0.970, collision 0.020, timeout 0.000, route completion 0.18, speed 5.60 m/s, energy 320.8 Wh per km. For comparison, clear: success 0.130, left the road 0.605, route completion 0.55.

8.6 What we can and cannot say

We can say	Because
A camera-only model trained this way is far below the grid-based models.	0.130 against 0.745–0.925 in clear weather, paired difference to the proposed model +0.795 (159 episodes where only the proposed model succeeded, none the other way).
Its main failure is leaving the road.	60.5% of clear episodes.
We cannot say	**Because**
That cameras cannot drive.	This is one small network, 40 plus 60 teacher episodes, no road or speed input.
That weather "breaks" it, if it already fails in clear weather.	Its clear-weather success is already near the floor; see the weather sentence above for what changes.

9 Reinforcement learning: what we tried (two-page record)

Reinforcement learning is not part of the five-model comparison. It is recorded here so the team knows what was tried. In all four attempts the algorithm is proximal policy optimisation (PPO), the action is the same two numbers (steering, throttle or brake, each between -1 and 1), and the network starts from an imitation-trained model, because a random start settles at standing still.

State (what the network is given). Attempt 1: 12 numbers of simulator state (six vehicle-state values, the previous action, a 4-number weather vector). Attempts 2 to 4: the bird's-eye-view grid and the vehicle-state numbers of the model they start from, plus (attempts 2 and 3) a weather or caution input.

Reward version 1 (attempts 1 to 3), per 0.1 s step:

$$r_t = \frac{\min(\Delta s,\, v_{cap}\Delta t)}{v_{ref}\Delta t} - 0.5\, d^2 - 5\, \psi^2 - 0.1\|a_t - a_{t-1}\|^2 - 0.002\, j^2 - 0.2\max(0, -a_t^{thr}) - 0.5\Big(\tfrac{(v-v_{cap})_+}{2}\Big)^2 - 0.2[v < 0.5] +$$

Δs distance driven, $v_{ref} = 10\,\mathrm{m/s}$, d sideways offset from the lane centre, ψ heading error, j jerk, a^{thr} throttle or brake command, v_{cap} speed cap (10 or 12 m/s), R_{end} terminal reward (+20 for success, -50 for a collision or leaving the road).

Reward version 3 (attempt 4):

$$r_t = \min\Big(\tfrac{\Delta s}{v_{ref}\Delta t}, \tfrac{v_{cap}}{v_{ref}}\Big) - 0.5d^2 - 5\psi^2 - 2\Big(1 - \tfrac{\tau}{3}\Big)_+^2 - \Big(1 - \tfrac{g}{6}\Big)_+^2 - 0.1\|a_t - a_{t-1}\|^2 - 0.002j^2 - 0.02E_t - 0.2[v < 0.5] + R_{end}$$

τ predicted time to collision (s), g gap to the vehicle ahead (m), E_t energy of the step (Wh); $R_{end} = +50$ for success and -100 for collision or leaving the road. The collision and gap terms use simulator truth during training only.

Table 10: The four reinforcement-learning attempts. "Validation" means the small set of episodes used during training; "test" means seeds never used for training.

Attempt	Start (initialisation)	Reward and what changed	Training (steps completed)	Testing	Result and lesson
1. Privileged teacher (runs C2, Probe C, C3)	Small network cloned from the stock driver on simulator state	Version 1, speed cap 12 (C2, Probe C) or 10 (C3), crash penalty 50; rain lowered tyre grip (later removed)	Learning rate $3 \cdot 10^{-4}$ ($5 \cdot 10^{-5}$ in C3), 512 steps per update, batch 512, 10 epochs. C2 1.5 M steps; Probe C 1 M; C3 0.3 M of 1.5 M	70 test episodes (10 clear, 10 fog, 50 rain); stock driver for reference	16 of 70 succeed (stock driver 47 of 70); all failures leave the road; reward rose while success did not. *Lesson:* a rising reward is not better driving.
2. Fine-tune the proposed model with a weather input	Proposed model; zero-initialised weather branch	Version 1, eight weather mixes with vehicle loss, density 0.1	Learning rate $5 \cdot 10^{-5}$, 256 steps per update, batch 128, clip 0.1. 120 000 of 480 000 steps	Validation on 40 episodes	0.675, 0.65, 0.675: no change. *Lesson:* a weather input cannot help a model that weather does not hurt.
3. Five reward combinations	Proposed model with zero-initialised caution input	Version 1 plus: none; energy term $-0.05E_t$; caution-dependent speed limit and crash penalty; headway term; all	As attempt 2 with 2 environments. 150 000 steps in each of five runs	Validation on 40 dense-traffic episodes	0.325–0.400 against a start of 0.350: noise. Energy did not fall. *Lesson:* 40 episodes cannot select a model; a reward cannot add missing information.
4. Reward version 3 under measured degradation	Weather-Degradation-Trained Driving Model	Version 3: adds collision-time and gap terms, no brake penalty, energy cost, $+50/-100$ terminal rewards	Learning rate 10^{-4}, 256 steps per update. 300 000 steps; the run with a weather input stopped after step 0	200 test episodes in four weather conditions	Success lower by 0.065, 0.115, 0.115, 0.095 (significant). Faster, less safe. *Lesson:* reward changes alone reduced safety; validation at 1.0 predicted nothing.

Overall: none of the four attempts beat the imitation-trained model it started from, and attempt 4 made it significantly worse. Training curves and before-and-after figures are in the technical report (reinforcement-learning section).

10 Which wording each result supports

Table 11: Claim, evidence and allowed wording.

Claim	Evidence	Allowed wording
The proposed model beats the baselines	200 paired episodes, $p = 0.0007$ and < 0.0001	"clearly better in sparse traffic in this simulator"
The grid matters	Image-only 0.130 against 0.925	"a camera-only model of this kind is far worse"
Weather does not hurt the proposed model	0.895–0.925 in every condition	"no measurable loss from vehicle loss at the measured rates in sparse traffic"; not "robust to weather"
The caution-aware model adapts	Slows down only when caution is high	"learned adaptive behaviour"; not "adaptive perception" or "senses weather"
The caution-aware model saves energy	35% lower estimate at lower speed	"lower estimated energy at lower speed"; not "more efficient"
Reinforcement learning did not help	Four attempts, one with 200 test episodes	"did not improve; once worsened"; not "cannot work"
The image-only model gets worse in weather	Success 0.130 in clear weather, then 0.000 (fog 1.0), 0.030 (rain 0.6), 0.010 (rain 1.0); 200 episodes each	That weather alone is the cause: it already fails 87% of clear-weather episodes, so weather only lowers an already low number

11 Where the files are

- Models: `trained_models/` (descriptive names; mapping in `trained_models/MODEL_NAME_MAPPING.md`). Models 1 and 2 are also on GitHub (`bev_policy/checkpoints/`).
- Code names (only here): model 1 `repo_bc_clear`, model 2 `repo_bc_sparse`, model 3 `ov_split`, model 4 `caut_a` (evaluated as `caut_auto`), model 5 `vcam`.
- Results: `bev_rl/adaptive/results/n200_all_weather/`, `bev_rl/adaptive/results/n200_horizon1600/`, `bev_rl/adaptive/results/vanilla_camera/`.
- Evaluation code: `bev_rl/adaptive/evaluate_n200_all_weather.py` (models 1 to 4), `bev_rl/adaptive/vanilla_camera_cnn.py` (model 5). Energy: `bev_rl/energy.py`.
- Figures and tables of this document: `report/make_five_model_comparison.py`, `report/make_detector_retention_figure.py`, `report/make_caution_longer_time_limit.py`, `report/make_five_model_160s.py`.